BRENDEN BUSKIRK

AI Deployment & Solutions Engineering — Selected Project Portfolio

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Three systems I designed, built, and validated. Each is here for what it shows about how I would run a customer AI deployment: find the decision that actually changes, build the smallest thing that changes it, and prove whether it worked.

> *Scope note — these are independently built systems, not commercial deployments. They are evidence of engineering judgment and validation discipline. My commercial deployment record is separate and documented: numerous capital-equipment deployments for recognizable enterprise manufacturers including OptumRx, Johnson Controls, Keysight Technologies, Samtec, and Graham Packaging — machine vision, AutoID, precision measurement, and laser displacement systems installed in live production environments.*

1 | Signal Validation & Real-Time Market Data Platform

Python · Polygon & Tastytrade APIs · OAuth · Statistical Validation

CHALLENGE The scanner was finding real directional signal — a measurable lift over a random baseline, enough to feel like an edge. Nothing in the pipeline tested whether that lift survived what it actually costs to trade it.

SYSTEM Multi-source ingestion and normalization across three market-data providers, real-time scanning and scoring, automated OAuth token refresh and session management, and an unattended daily operating cycle with failure sentinels.

EVIDENCE I built the layer that audited my own system: random-baseline comparison, bootstrap confidence intervals, Cohen's d, and execution-aware Black-Scholes modeling. The directional lift over random held up. The economics did not — approximately -10.9% modeled option return, roughly a 10% win rate, Kelly fraction 0.00%.

WHAT IT SHOWS A system can predict something real and still fail to produce an economically useful decision. I instrument the thing that tells the difference, and I report the result when it is the one I did not want. That is why an AI pilot survives its first quarterly review instead of quietly not being renewed.

2 | FR8 Boss — Freight Decision-Support Engine

Python · CLI Workflows · YAML Scenarios · Business Rules · Explainable Output

CHALLENGE Load decisions get made in minutes across deadhead, timing, capacity, insurance, backhaul, and margin. Judgment varies by dispatcher and by hour, and the losses stay invisible because nobody logs the reasoning.

SYSTEM A decision-support workspace that evaluates load economics against repositioning cost, revenue per mile, capacity, insurance constraints, backhaul alternatives, time decay, and decision deadlines, driven by YAML scenarios.

EVIDENCE Converts unstructured operating inputs into PASS / WARN / INTERVENE recommendations with the reasoning written out and a reusable audit record. The operator keeps the final call.

WHAT IT SHOWS Map the real workflow before automating it. Make the machine's reasoning inspectable. Keep the human in the loop where context beats rules — and leave behind an audit trail the customer can inspect.

3 | Mulligan Made — Computer-Vision Swing Analysis

MediaPipe · OpenCV · Signal Processing · FFmpeg · Media & PDF Generation

CHALLENGE A working pose model gets you maybe a third of the way. The distance between correct motion data and guidance a customer can act on is most of the remaining work.

SYSTEM End-to-end pipeline: swing-phase detection, pose-data smoothing, tempo and biomechanics measurement, then translation of motion data into plain-language guidance.

EVIDENCE Produces annotated replay video, motion overlays, heatmaps, PDF summaries, and QR-linked deliverables — upload to finished artifact with no manual steps in between.

WHAT IT SHOWS Model output is not the deliverable. The last mile — turning a correct result into something the customer immediately understands — is where deployments succeed or fail.

HOW I APPROACH AI DEPLOYMENT

Find the decision that actually changes · Define success before building · Build the smallest useful implementation · Instrument the business outcome, not the model metric · Keep humans in control where judgment matters · Turn what worked into a repeatable playbook